Industrial Safety Processes

Powered Industrial Trucks (PITs)

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Generally Classified by:

- Power source
- Operator position
- Means of engaging loads

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Forklifts

- Most common form of powered industrial truck
- Come in several variations and varieties
- Differences can cause problems with:
  - Operator competency
  - Safe operation in different environments

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Variations in Forklifts – Power Source

- Gasoline
- Diesel
- Propane
- Electric

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Size and Maximum Loads

- Maximum weight of loads
- Maximum height of lifts
- Weight of truck itself

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Variations in Forklifts

- Nature of attachments and material handling
  - Sometimes a “forklift” in name only
- Rear or front wheel turning control
- Normal or hazardous locations

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Numerous Attachment Options

- Standard forklifts with varying lengths of forks
- Paper or roll handling
- Drum picking and transport
- Carpet or long roll transport
- Lifting hooks
- Snowplow
- Man basket lift

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General Considerations

- Controlling exhaust
- Protecting infrastructure – racking, utilities
- Protecting the operator – cages
- Danger areas – loading docks, slopes, blind corners
- Pedestrians and other equipment

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Protecting Operators

- Use seatbelts
- Protective cages

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Pedestrians & Other Equipment

- Separate activities as much as possible
- Lighting & backup beeps
- Establish firm right-of-way policy – and enforce it

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Key Issues in Forklift Driving

- Forklifts are generally steered by their rear wheels
- Forklifts tend to steer more easily when carrying a load
- Forklifts are usually driven in reverse as much as forward
- Forklifts are often steered with only one hand

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Safe Operation

- Slow/stop at intersections or blind corners
- Honk at intersections, blind corners, pedestrians
- Speed limits – indoors and out
- Maintaining visibility over loads
- Look where you are going

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Leaving the Forklift

When leaving a forklift, the operator should:

- Take the unit out of gear (put in neutral)
- Shut off the power
- Set the brake,
- Remove the key
- Remove any connector plug
- Set the load-engaging mechanism in lowered or inoperative position

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Forklift Stability Triangle

- Forklifts have a “stability triangle” that impacts the loads they can lift safely
- The center of gravity (CG) must be within the triangle for the forklift to operate safely
- Lifting or position loads so that the center of gravity (CG) moves outside the triangle may result in an accident

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Stability Triangle & Grades

- Need to be careful moving up and down slopes, such as ramps
- When moving up and down ramps more than 10% grade:
  - Loaded forklifts should drive with the load toward the upgrade
  - Unloaded trucks should drive with the forks toward the downgrade

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Inspections

- Pre-use usually required by MOM
- Company should enforce, especially if there are multiple shifts or users of PITs
- Should be basic and not include any repairs
  - Lights, brakes, alarms / beeps, smooth operation, etc.

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Maintenance & Repairs

- Regular, scheduled
- 3rd party recommended if expertise is not in-house
- Follow MOM at a minimum

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Operator Training & Reinforcement

- Proper training of operators
- Internal “rodeos” as deemed necessary
- Monitoring performance and correcting deficiencies

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Operator Training

- Include detailed classroom instruction on safe operation and inspection of forklifts
- Include detailed field testing of operator competence in controlled setting
- Include ongoing refresher instruction and review of operator performance
- All training and monitoring should be clearly documented

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Main Forklift Safety Issues

- Operator horseplay
- Operator error
- Operator not using or disabling safety devices
  - Seat belt, beeps (motion or back), horn at corners
- Speed
- Lack of inspection and maintenance
- Improper loading

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Other Industrial Trucks

- Straddle Trucks
- Automated Guided Vehicles (AGVs)
- Crane Trucks (Mobile Cranes)
- Yard Tractors
- Rail Tractors/Car Movers
- Motorized Hand Trucks
- Order Pickers

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Other Industrial Trucks

- Same basic safety issues apply as forklift trucks
- Operator training and monitoring is critical
- Each type of truck has its own special issues that need to be addressed
- Training with one type of truck is generally not transferrable to others

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Straddle Trucks

- Used to transport long loads
  - Lumber, piping
- Often driver has restricted visibility
  - Cannot see everything in front or behind
  - Potential danger to pedestrians

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Automated Guided Vehicles (AGV)

- May operate based on several methods of instruction
  - Under-floor wiring, paint coloring, chemical tracing, radio/computer direction
- Concern with overloading
- Concern with moving over or across slopes
- Need additional safety features
  - Soft-stop bumpers, near contact sensing

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Yard Tractors

- Often used to move trailers in yards
- Can be of various sizes and power configurations
  - Operator may require same licensing as tractor trailer operator
- Represent common large vehicle hazard
- Visibility is not always very good

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Rail Tractors / Car Movers

- Used to shuttle rail cars from location to location
- Various power sources – typically diesel
- Rail operation – typically not great for sudden breaking

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Motorized Hand Trucks

- Can be easier to operate than forklifts
- Subject to many of the same problems as forklift use
  - Speed, horseplay
- Subject to many of the same problems as regular hand trucks
  - Visibility, crushing hands

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Order Pickers

- Combine hazards of industrial truck with working at heights
- Often operators are not properly trained on fall protection systems normally required by these systems
- Operator error is a major problem with order pickers

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